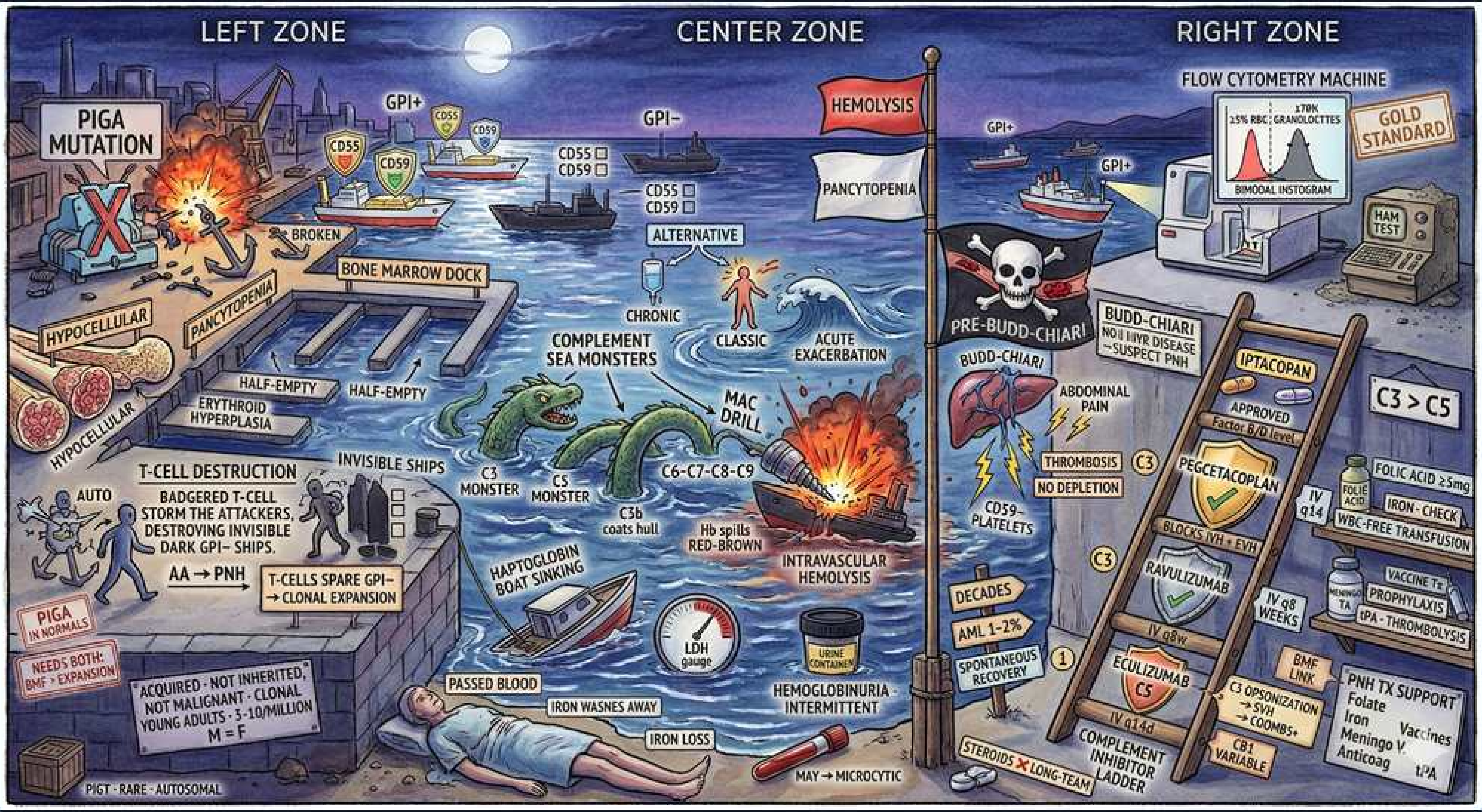
Paroxysmal Nocturnal Hemoglobinuria (PNH)



1. PIGA mutation

WHAT YOU SEE

Left zone, PIGA mutation, X

WHAT IT ENCODES

PNH arises from an ACQUIRED somatic mutation in the PIGA gene in a hematopoietic stem cell. PIGA is needed to make the GPI anchor; without it, GPI-anchored surface proteins (CD55, CD59) are missing. It is clonal, not inherited.

BOARD TRAP

Calling PNH inherited; it is an acquired clonal somatic PIGA mutation, X-linked gene but not germline-transmitted.

2. Missing CD55/CD59

WHAT YOU SEE

CD55/CD59 absent flags

WHAT IT ENCODES

GPI anchor loss removes complement regulators CD55 (DAF) and CD59 (MIRL) from blood cells. Without them, the membrane attack complex lyses RBCs unchecked. CD59 deficiency is the main driver of hemolysis.

BOARD TRAP

Attributing the hemolysis to autoantibodies; PNH hemolysis is complement-mediated from missing CD55/CD59, DAT is NEGATIVE.

3. Complement-mediated lysis

WHAT YOU SEE

Hemolysis flag, complement sea monsters

WHAT IT ENCODES

Uncontrolled complement (MAC) causes chronic INTRAvascular hemolysis: low haptoglobin, high LDH, hemoglobinuria, hemosiderinuria. Free hemoglobin scavenges nitric oxide, causing smooth-muscle dystonia (esophageal spasm, abdominal pain, erectile dysfunction).

BOARD TRAP

Expecting extravascular/splenic hemolysis; PNH is complement-mediated INTRAvascular hemolysis with hemoglobinuria.

4. Nocturnal hemoglobinuria

WHAT YOU SEE

Red-brown urine, intermittent

WHAT IT ENCODES

Classic but inconstant: dark red-brown morning urine (nocturnal acidosis enhances complement). Hemoglobinuria is intermittent. Iron is lost in urine, so chronic PNH can paradoxically cause iron DEFICIENCY despite ongoing hemolysis.

BOARD TRAP

Requiring nocturnal hemoglobinuria for diagnosis; it is present in a minority — most PNH lacks the classic dark morning urine.

5. Pancytopenia / marrow failure

WHAT YOU SEE

Pancytopenia flag, bone marrow duck, hypocellular

WHAT IT ENCODES

PNH overlaps with aplastic anemia — many patients have a hypocellular marrow and pancytopenia. The PNH clone expands in the setting of immune marrow failure ('escape'). Screen aplastic anemia/MDS patients for a PNH clone.

BOARD TRAP

Treating PNH as purely a hemolytic disease; bone marrow failure/aplastic anemia coexists and often dominates the picture.

6. Thrombosis

WHAT YOU SEE

Right zone, thrombosis lightning, Budd-Chiari

WHAT IT ENCODES

THROMBOSIS is the leading cause of death in PNH — characteristically at unusual sites (hepatic vein = Budd-Chiari, cerebral, splanchnic, dermal). Complement-activated platelets and free Hb drive it. Unexplained intra-abdominal/hepatic vein clot + hemolysis = think PNH.

BOARD TRAP

Overlooking PNH in a young patient with Budd-Chiari; PNH thrombosis classically hits the hepatic and other unusual veins.

7. Flow cytometry diagnosis

WHAT YOU SEE

Flow cytometry machine, gold standard

WHAT IT ENCODES

Flow cytometry is the GOLD STANDARD: detects loss of GPI-anchored proteins (CD55, CD59) on RBCs and WBCs, and FLAER (fluorescent aerolysin) binds the GPI anchor on granulocytes/monocytes. Quantifies clone size. Replaces the old Ham/sucrose tests.

BOARD TRAP

Ordering the Ham acid-hemolysis or sucrose lysis test; flow cytometry with FLAER is now the standard diagnostic.

8. FLAER on granulocytes

WHAT YOU SEE

Bimodal histogram, granulocytes

WHAT IT ENCODES

FLAER testing on granulocytes/monocytes most reliably sizes the PNH clone (RBC fraction underestimates it because hemolyzed PNH RBCs are destroyed). Clone size guides treatment and prognosis. Bimodal histogram = normal + deficient populations.

BOARD TRAP

Sizing the clone on RBCs alone; hemolysis depletes PNH RBCs, so granulocyte FLAER better reflects true clone size.

9. Eculizumab (C5 inhibitor)

WHAT YOU SEE
Ladder — eculizumab, C5

WHAT IT ENCODES
Eculizumab/ravulizumab are anti-C5 monoclonal antibodies that block terminal complement (MAC), reducing intravascular hemolysis, transfusion need, and thrombosis. Mainstay for hemolytic/thrombotic PNH. Ravulizumab is longer-acting.

BOARD TRAP
Expecting C5 inhibitors to fix the marrow; they control complement hemolysis/thrombosis but don't treat marrow failure.

10. Meningococcal vaccination

WHAT YOU SEE
Vaccine, meningococcus warning

WHAT IT ENCODES
C5 inhibitors block terminal complement, sharply raising risk of Neisseria meningitidis infection. Patients MUST receive meningococcal vaccination (and often penicillin prophylaxis) before starting eculizumab/ravulizumab. This is a high-yield safety point.

BOARD TRAP
Starting eculizumab without meningococcal vaccination; terminal complement blockade predisposes to lethal meningococcal sepsis.

11. C3 > C5 extravascular shift

WHAT YOU SEE
C3 > C5 sign

WHAT IT ENCODES
On C5 inhibitors, surviving PNH RBCs accumulate C3 fragments and may be cleared EXTRAvascularly by the liver/spleen, causing residual anemia despite blocked intravascular hemolysis. Newer C3/proximal inhibitors (pegcetacoplan) address this.

BOARD TRAP
Assuming persistent anemia on eculizumab means failure; it can be C3-mediated extravascular hemolysis, treated with proximal (C3) inhibitors.

12. Haptoglobin / LDH labs

WHAT YOU SEE
Haptoglobin boat sinking, LDH shop

WHAT IT ENCODES
PNH labs: very low haptoglobin, markedly high LDH, reticulocytosis, hemoglobinuria/hemosiderinuria, and a NEGATIVE direct antiglobulin test. The DAT-negative intravascular hemolysis pattern is a key PNH clue.

BOARD TRAP
Expecting a positive DAT; PNH hemolysis is complement-mediated, not antibody-coated — the DAT is negative.

13. Allogeneic transplant

WHAT YOU SEE
Ladder bottom, BMT, var/cure

WHAT IT ENCODES
Allogeneic hematopoietic stem cell transplant is the only curative therapy and is reserved for severe marrow failure or refractory disease, given its toxicity. C5 inhibition has reduced the need for transplant in hemolytic PNH.

BOARD TRAP
Offering transplant for isolated hemolytic PNH; complement inhibition is preferred — transplant is for severe marrow failure.

14. AML/MDS evolution

WHAT YOU SEE
AML/MDS lightning

WHAT IT ENCODES
PNH carries a small risk of clonal evolution to myelodysplastic syndrome or acute myeloid leukemia, reflecting its origin in a damaged stem cell compartment. Long-term hematologic monitoring is required.

BOARD TRAP
Treating PNH as static; it can evolve to MDS/AML, so ongoing marrow surveillance is warranted.

15. Acquired not inherited tag

WHAT YOU SEE
Banner — not inherited, M=F

WHAT IT ENCODES
PNH is acquired (somatic), affects men and women equally, and is not malignant per se but is a clonal stem-cell disorder. Typically presents in young to middle-aged adults with hemolysis, thrombosis, and/or cytopenias.

BOARD TRAP
Screening family members; PNH is a somatic acquired clone with no Mendelian inheritance to trace.

16. Chronic + acute exacerbation

WHAT YOU SEE
Center, chronic vs acute exacerbation

WHAT IT ENCODES
Baseline chronic low-grade intravascular hemolysis is punctuated by acute exacerbations triggered by infection, surgery, or stress (complement amplification). Recognizing triggers helps anticipate hemoglobinuria crises and transfusion needs.

BOARD TRAP
Assuming stable hemolysis; infection/surgery amplify complement and provoke acute PNH hemolytic crises.